

Dalron J. Robertson, M.S.

Research Scientist, Pharm.D./MBA Candidate

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Education

Chicago State University – *Doctor of Pharmacy, Expected May '28*

Mississippi College – *MBA, Expected May '28*

Mississippi College – *M.S., Biological Sciences*

University of Illinois, Chicago – *Software Engineering Certificate*

Jackson State University – *B.S., Biology*

Technical Skills

AI & Large Language Models: Deepseek | FAISS | Gemini | LLaMA | LoRA | OpenAI | TinyLlama | TxGemma

Bioinformatics & Molecular Modeling: AlphaFold | ChimeraX | SwissTargetPrediction | Molinspiration | RDKit | UniProt

DataScience & Machine Learning: Tensorflow | Scikit-Learn | Pandas | NumPy | PyTorch

Languages: Javascript | Python | Pinescript | Java | C | C++ | PostgreSQL | MySQL

Tools & Technologies: Git | AWS | ISOLIDE | | Schrödinger | jest | DocuTrack | Framework LTC | UiPath

Web Development: Node.js | React.js | Express.js | FastAPI | Flask | HTML | CSS | AJAX | DOM | jQuery

Experience

UChicago Medicine, Hyde Park, Chicago, IL

Pharmacy Intern – Discharge Medication Program | Mar 2026 -

- Support all operational roles across the Discharge Medication Program and outpatient pharmacy, preparing and dispensing prescriptions, delivering medications to the bedside, and counseling patients before discharge.
- Prepare a broad range of nonsterile compounded medications across oral and topical dosage forms, following established formulas, calculations, documentation, and quality procedures.
- Operate Parata Max automated dispensing systems through prescription filling, stocking, replenishment, and troubleshooting; receive and reconcile pharmacy inventory and assist with medication ordering.
- Coordinate medication access by conducting insurance and benefits investigations, completing and tracking prior authorizations, facilitating patient-assistance enrollment, and partnering with pharmacists, providers, clinics, case management, and social services to prevent therapy interruptions.
- Document access activity in pharmacy tracking systems, manage inbound and outbound patient calls, resolve coverage barriers with insurers, and identify workflow opportunities that improve reimbursement and patient service.

Astellas Pharmaceuticals, Northbrook, IL

Mentee – Bridge to Industry Intensive Program | Oct 2025 -

- Selected participant in a professional development partnership between Chicago State University College of Pharmacy and Astellas Pharma Global Development.
- Engage in one-on-one mentorship with Astellas executives, gaining exposure to Quality, Research, and Regulatory leadership.
- Participate in seminars and applied learning sessions focused on pharmaceutical innovation, compliance, and global development pathways.

Walgreens, Chicago, IL

Pharmacy Intern | Oct 2025 - Jul 2026

- Support pharmacists in medication verification, dispensing workflows, and clinical decision-making processes in accordance with state and federal pharmacy regulations.
- Interpret patient profiles, medication histories, and prescription data within pharmacy systems to ensure accuracy, appropriateness, and regulatory compliance prior to dispensing.

Mississippi College, Clinton, MS

Lead Graduate Researcher | Dec 2024 - Jul 2025

- Applied AI-driven literature mining and bioinformatics to extract and synthesize bioactivity, phytochemical, and pharmacological data from large biomedical databases.
- Built and analyzed a curated library of *Nigella sativa* constituents, performing molecular descriptor analysis, drug-likeness screening, and lead prioritization.
- Identified and modeled protein targets using SwissTargetPrediction and AlphaFold, and conducted molecular docking and visualization to assess ligand-target interactions.

Scale AI, San Francisco, CA

AI Software Engineer | Sept 2023 - Jan 2025

- Collaborated with Google teams to improve Gemini and Google Search models by developing high-quality training datasets that enhanced accuracy, relevance, and performance.
- Analyzed large-scale model outputs and built synthetic data pipelines to improve temporal and relational reasoning, contributing to measurable gains in model precision.

Medication Management Partners, Crestwood, IL

Pharmacy Informatics & Automation, Pharmacy Intern | Aug 2020 - Apr 2022

CVS Health, Chicago, IL

Pharmacy Intern | Jul 2015 - Aug 2020

Research & Development Projects

NaS Knowledge Model | June 2025 | <https://github.com/NaS-Research/knowledge-model>

Engineered a biomedical retrieval-augmented LLM based on Google Deepmind's TxGemma 2B. The model utilizes LoRA fine-tuning to produce literature-supported responses and autonomously updates nightly with newly ingested research data.

Key Achievements & Technical Highlights:

- Reduced model training from multiple days to less than four hours using targeted LoRA adapters (error reduced from 5.16 → 2.23 within ~300 steps).
- Increased retrieval accuracy through 1,024-dimensional BGE-large embeddings and a two-stage retrieval strategy.
- Automated daily ingestion, embedding, and indexing of biomedical data, eliminating manual intervention.
- Implemented rigorous context validation, decreasing unsupported claims by approximately ~34%.

Phytochemical Bioactivity Research | Jan 2025 - May 2025 | <https://bit.ly/DIR-PhytochemicalResearch>

Conducted computational and experimental research on ethanol-derived *Nigella sativa* extracts using bioinformatics, molecular modeling, and AI-driven analysis. Presented findings at symposium; recognized by Provost for research excellence.

- **Phase 1: Literature Mining & Data Collection**
 - Leveraged Deep Research methodologies (OpenAI's LLMs) to rapidly extract and synthesize bioactivity, pharmacological, and chemical data, from PubMed, PubChem, and UniProt databases.
- **Phase 2: Compound Library & Chemical Profiling**
 - Developed a comprehensive chemical library of *N. sativa* phytochemicals, conducting detailed molecular descriptor calculations, ADME/Tox assessments, and archiving structural data for computational analyses.
- **Phase 3: Target Identification & Structure Modeling**
 - Identified protein targets via SwissTargetPrediction; validated through extensive literature reviews. Generated 3D protein models using AlphaFold; developed visualizations with ChimeraX and Blender.
- **Phase 4: Virtual Screening & Molecular Docking (Current Phase)**
 - Performing molecular docking simulations to predict binding affinities and interaction dynamics between prioritized phytochemicals and protein targets.
 - **Teaching & Mentorship:** Trained and mentored 16 undergraduate and graduate researchers on the complete computational pipeline, including software and analytical methods, significantly enhancing research productivity.